

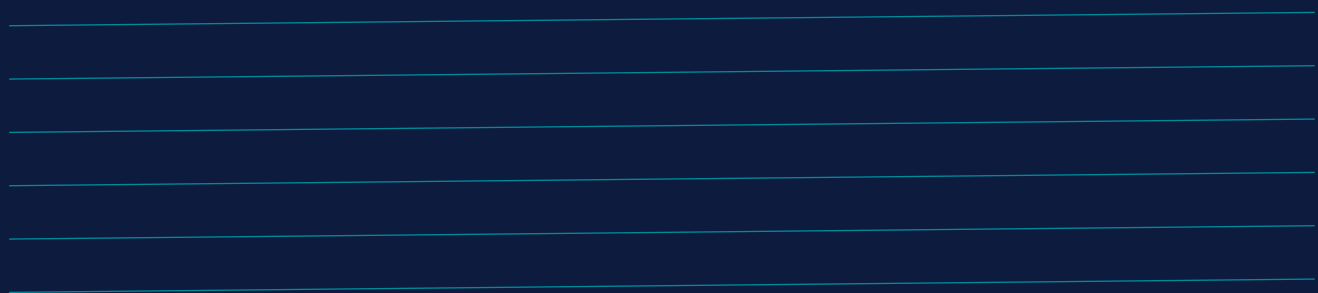
QBITEL BRIDGE

AVIATION & AEROSPACE SECURITY PLATFORM

Quantum-Safe Air Traffic and Avionics Protection

ADS-B Authentication | Bandwidth-Optimized PQC | DO-326A Compliance
ARINC 653 Security Partition | Ground-Only Deployment

The safest industry in the world has no authentication on its primary surveillance system. QBITEL Bridge changes that.



DO-326A Evidence Ready

180K+

Daily Unauthenticated ADS-B
Messages

\$20

Cost to Spoof
Aircraft Position

60-80%

PQC Signature Compression

Zero

Aircraft Changes Required

EXECUTIVE SUMMARY

The Safest Industry in the World Has No Authentication on Its Primary Surveillance System

Aviation has achieved an extraordinary safety record through decades of rigorous engineering, redundant systems, independent oversight, and a culture of relentless improvement. Yet this same industry — which mandates triple-redundant hydraulics and requires independent verification of every line of flight-critical software — broadcasts the precise position of every aircraft on earth over an open radio channel with no authentication, no encryption, and no mechanism to distinguish a real aircraft from a spoofed one.

ADS-B (Automatic Dependent Surveillance-Broadcast) is the backbone of modern air traffic surveillance, mandated by the FAA, EASA, ICAO, and virtually every civil aviation authority worldwide. It was designed as an open broadcast system in the 1990s and 2000s when cryptographic authentication of 1090 MHz broadcasts was considered impractical. Today, a \$20 software-defined radio is all that is required to inject ghost aircraft into the global air traffic picture.

180K+

Daily ADS-B Messages
— Zero Authentication

\$20

Cost to Spoof
Aircraft Position with SDR

60-80%

PQC Signature Compression
Achieved

36 Wks

Typical Full Deployment
Timeline

What QBITEL Bridge Delivers

Cryptographic ADS-B Authentication — Ground-receiver-level authentication with zero aircraft modifications or DO-178C recertification.

Bandwidth-Optimized PQC — Post-quantum cryptography compressed 60-80% to fit LDACS (600 bps) and ACARS (2.4 kbps) constraints.

AI-Driven Spoofing Detection — Multilateration cross-validation detects ghost aircraft within 10 seconds of injection.

ARINC 653 Security Partition — Isolated airborne security partition for IMA platforms — DAL-D certified, no safety partition impact.

DO-326A/DO-178C Evidence Generation — Complete Secure Development Assurance documentation accepted by FAA and EASA.

ATC Network Quantum Hardening — Quantum-safe VPN mesh protecting EUROCONTROL, NATS, FAA SWIM infrastructure.



NIST finalized PQC standards August 2024. U.S. government mandates PQC migration by 2035. Aircraft certified today will operate until 2060-2065. Act now.

THE THREE CRITICAL THREATS

ADS-B Spoofing | Bandwidth-Constrained Quantum Exposure | Certification Barrier

Threat 1: ADS-B Spoofing and Ghost Aircraft

ADS-B OUT requires every aircraft above certain altitudes to continuously broadcast its ICAO 24-bit address, callsign, GPS position, altitude, velocity, and emergency status on 1090 MHz — a frequency accessible to any radio receiver. There is no authentication field in the ADS-B message format. Any transmitter broadcasting correctly formatted messages will be accepted by ground stations and displayed on ATC radar screens.

Attack Vector

- Monitor legitimate ADS-B transmissions to understand traffic patterns
- Generate spoofed ADS-B messages for non-existent ghost aircraft (\$20 SDR + open-source software)
- Inject false position updates for real aircraft — position falsification
- Create phantom traffic that triggers TCAS resolution advisories in real aircraft
- Gradually shift an aircraft's apparent position to induce air traffic controller errors

Threat Vector	Attack Cost	Current Detectability	QBTEL Response
ADS-B ghost aircraft injection	\$20 SDR	None — no authentication	Detection <10 sec
ADS-B position falsification	\$20 SDR	None	Detection <30 sec
ACARS message forgery	\$500 SDR+decoder	None	Real-time PQC auth
CPDLC clearance spoofing	Sophisticated	Limited	PQC + session binding
Harvest-now-decrypt-later	Passive collection	None	PQC eliminates threat

Threat 2: Bandwidth-Constrained Quantum Exposure

Standard post-quantum signatures are enormous by aviation standards. A raw ML-DSA-65 (Dilithium-3) signature is 3,309 bytes. At 2,400 bps ACARS, this requires 11 seconds to transmit — an eternity for time-sensitive operational communications. Aviation data links operate at bandwidths that would seem impossibly restrictive to enterprise IT professionals.

Data Link	Bandwidth	Primary Use	Raw PQC Overhead	Compressed Overhead
LDACS (next-gen)	600 bps–2.4 kbps	Air-ground digital	Too large	<15% Falcon-512
VHF ACARS	2,400 bps	Operational comms	1,100%	<8% session cache
HF ACARS	300–1,800 bps	Oceanic comms	Too large	<12% session-based
SATCOM Iridium	600 bps–128 kbps	Global coverage	1,100%	<8% with caching

SATCOM Inmarsat	1.2–432 kbps	Broadband ops	252%	<3% full PQC
ATC Networks	Gbps Ethernet	Ground systems	<0.1%	Not required

QBITEL Bridge achieves 60-80% signature compression through algorithm-specific lossless compression exploiting lattice signature structure, context-aware delta encoding, session key amortization, hierarchical signing with time-bounded session certificates, and selective field authentication. Compressed Falcon-512 fits LDACS frame constraints. Compressed Dilithium-3 adds less than 0.8 seconds latency on 2,400 bps ACARS.

Threat 3: The DO-178C Certification Barrier

Aviation software certification under DO-178C is the most rigorous software quality process in any industry. For Design Assurance Level A (catastrophic failure conditions), DO-178C requires full requirements traceability, Modified Condition/Decision Coverage (MC/DC) for every decision point, independent verification of every test procedure, and comprehensive documentation of every design decision. The cost of DO-178C Level A certification for a new avionics function ranges from \$50M to \$500M per aircraft type.

This creates a security paradox: the most safety-conscious industry in the world is structurally prevented from rapidly deploying security updates. Boeing 737 MAX aircraft delivered in 2024 will likely still be in service in 2055-2060. The cryptographic algorithms protecting their communications must remain secure for their entire operational life.

QBITEL's Response: The primary deployment architecture is **ground-only** — no changes to certified airborne software, eliminating the DO-178C recertification requirement entirely. For airborne deployment, QBITEL provides a DAL-D ARINC 653 security partition with complete DO-178C evidence generation.

SEVEN CORE CAPABILITIES

ADS-B Auth | Bandwidth PQC | LDACS/ACARS/SATCOM | ARINC 653 | Ground-Only | DO-326A | ATC Networks

Capability 1: ADS-B Authentication and Anti-Spoofing

Layer 1 — ICAO Address Binding: Aircraft ICAO 24-bit addresses cryptographically bound to operator certificates in distributed aviation PKI.

Layer 2 — Message Authentication Codes: MACs computed and verified at ground stations via transponder-adjacent hardware — no airborne software certification.

Layer 3 — Behavioral Authentication: AI engine performs behavioral authentication for non-participating aircraft, verifying position consistency with performance envelopes.

Layer 4 — Cross-Source Validation: ADS-B cross-validated against SSR Mode S radar, MLAT calculations, and historical flight profile data.

Value	Metric	Value				
<50 ms	Throughput	500K+ msg/hr	False Positive	<0.001%	Detect Rate	>99.7%

Capability 2: Bandwidth-Optimized PQC for Constrained Data Links

Algorithm-Specific Compression: Exploit mathematical structure of lattice signatures for 60-80% lossless compression unique to each algorithm.

Context-Aware Delta Encoding: Aviation messages follow predictable patterns; QBITEL transmits cryptographic deltas rather than full signatures.

Session Key Amortization: Single expensive PQC handshake; subsequent messages use lightweight symmetric MACs on the same link.

Hierarchical Signing: Ground infrastructure signs time-bounded session certificates for rapid per-message authentication without per-message PQC overhead.

Value	Metric	Value				
60-80%	Falcon-512 Size	359-538B	ACARS Overhead	<0.8 sec	LDACS Compliance	Yes

Capability 3: LDACS / ACARS / SATCOM Security

LDACS (Next-Gen Air-Ground): Quantum-safe key establishment, CPDLC message authentication, ATC clearance protection, privacy for L-band transmissions.

VHF/HF ACARS (Legacy): Compressed PQC authentication, end-to-end integrity, non-repudiation for maintenance write-ups and MEL deferrals.

SATCOM (Iridium and Inmarsat): Bandwidth-adaptive PQC selection, hybrid encryption for VoIP SATCOM, ACARS-over-SATCOM (AOS) stream protection.

CPDLC Protection: Controller-Pilot Data Link Communications secured against clearance spoofing with session-bound PQC authentication.

Value	Metric	Value				
6+	Bandwidth Adapt	Yes	AOS Support	Yes	VoIP Encrypt	Yes

Capability 4: ARINC 653 Security Partition (Airborne)

Partition Isolation: Zero shared memory with flight-critical partitions. Strict time allocation cannot delay flight-critical processing.

Partition Functions: Local key storage, PQC computation, authentication verification, security event logging, certificate management.

DO-178C DAL-D Certification: Partition failure cannot propagate to safety-critical systems due to ARINC 653 isolation guarantees.

Evidence Package: Complete HLR/LLR, Software Architecture, source code traceability, unit tests, MC/DC coverage, tool qualification.

Value	Metric	Value				
787, A380, A350	DAL Level	DAL-D	Isolation	ARINC 653	Cert Evidence	Full DO

Capability 5: Ground-System-Only Deployment

No Aircraft Changes: ADS-B spoofing attacks occur at the ground receiver. Authentication at the ground receiver intercepts spoofed messages without aircraft involvement.

No DO-178C Recertification: Ground-only deployment triggers no airborne software change — the most expensive and time-consuming aspect of aviation security.

Complete Coverage: ADS-B authentication, ATC network security, ACARS ground side, SATCOM ground side, airport surface surveillance — all without aircraft modification.

Immediate Deployment: Ground-only deployment can be initiated within weeks. No aircraft access, maintenance slots, or airworthiness approvals required.

Value	Metric	Value				
None	Recertification	None	Deploy Time	Weeks	ADS-B Coverage	Full

Capability 6: DO-326A / DO-178C Compliance Evidence Generation

DO-326A Documentation: Aircraft Security Log (ASL), Security Target Analysis (STA), Security Development Analysis (SDA), Security Assessment Report.

DO-178C Evidence (Airborne): Software Plans, Development artifacts (HLR/LLR, Architecture, Design, Source Code), Verification (tests, results, MC/DC coverage).

FAA and EASA Alignment: Evidence structured for FAA Aircraft Certification Service Issue Papers and EASA Certification Review Items (CRIs).

Ongoing Compliance: Continuous evidence generation throughout system lifecycle. Configuration management integration ensures evidence currency.

Value	Metric	Value				
Full ASL/STA/SDA	DO-178C Evidence	Complete	FAA Alignment	Yes	EASA Alignment	Yes

Capability 7: ATC Network Quantum Hardening

SWIM Network Protection: Quantum-safe VPN mesh for FAA SWIM NAS and EUROCONTROL SWIM B2B APIs — securing NextGen/SESAR information sharing.

Inter-Facility Communications: Quantum-safe encryption for AIDC (ATS Inter-facility Data Communications) and OLDI (On-Line Data Interchange) links.

AFTN/AMHS Gateway Security: Quantum-safe wrapping for Aeronautical Fixed Telecommunication Network and Aeronautical Message Handling System traffic.

HSM and SIEM Integration: Hardware Security Modules for ATC cryptographic key storage. SIEM integration with aviation-specific threat intelligence feeds.

Value	Metric	Value				
ACC, TRACON, Tower	HSM Integration	Yes	SIEM Integration	Yes	SWIM Support	Yes

COMPLIANCE COVERAGE

DO-326A | DO-178C | FAA AC 119-1 | EASA CS-STAN | ICAO Annex 10 | NIST FIPS 203/204/205

QBITEL Bridge provides complete regulatory compliance coverage for aviation security certification across all major jurisdictions and standards bodies. The following table maps each applicable standard to QBITEL's coverage scope and the specific evidence artifacts generated.

Regulation / Standard	Scope	QBITEL Coverage	Evidence Provided
DO-326A (ED-202A)	Airworthiness Security Process	Full	ASL, STA, SDA, SAR
DO-356A (ED-203A)	Security Methods and Considerations	Full	Method selection docs
DO-178C (ED-12C)	Airborne Software Quality	Partition-scoped	Full Plan/Dev/Verify
DO-254 (ED-80)	Airborne Electronic Hardware	Interface documentation	HW-SW interface docs
FAA AC 119-1	Aircraft Network Security Program	Full	ANSP documentation
EASA CS-STAN	Standard Changes and Repairs	Applicable standards	Change justification
ICAO Annex 10	Aeronautical Telecommunications	Alignment	Technical compliance
EUROCAE ED-205	Aviation Wireless Cyber Security	Full	Compliance mapping
RTCA SC-216 / DO-260C	ADS-B MOPS	Security extensions	Technical interface doc
NIST FIPS 203 / 204 / 205	Post-Quantum Cryptography Stds	Full implementation	Algorithm cert docs
NIST SP 800-208	PQC Key Management	Full	Key management procedures
Common Criteria EAL4+	Security Product Evaluation	Full CC package	CC evaluation docs

C All compliance documentation is generated automatically by BRIDGE-AV-CERT and is formatted for direct submission to FAA, EASA, CAAC, and national aviation authorities.

INTEGRATION ECOSYSTEM

Honeywell | Thales | Collins Aerospace | SITA | NATS | FAA/EUROCONTROL

Avionics OEM Partners

Honeywell Aerospace: Connected Aircraft platform integration. APEX IMA platform compatibility. Primus avionics suite integration. HTR system interfaces for spoofing alert correlation.

Thales Avionics: IMA platform (TopTech) security partition integration. FLYSMART+ operational support system security. AVMS network protection. TCAS/ACAS integration for spoofing alerts.

Collins Aerospace: Pro Line Fusion avionics suite integration. GLOBALink SATCOM security enhancement. MultiScan weather radar data link protection. ARINC 429 interface security gateway.

Air Navigation Service Providers

NATS (UK): NERC interface and iCAS integration. Swanwick and Prestwick Area Control Centre deployment support. NERC engineering coordination for receiver network authentication.

EUROCONTROL: SWIM Yellow Profile security extension. B2B API quantum protection. CFMU network security and Network Manager interface.

FAA (USA): SWIM NAS integration for NextGen. TFMS (Traffic Flow Management System) network protection. STARS (Standard Terminal Automation Replacement System) security overlay.

Communication and Data Service Providers

SITA: ACARS quantum-safe upgrade path for AviNet. Global aviation network security for WorldTracer and baggage system infrastructure. Airport community systems protection.

ARINC (Collins Aerospace): GlobalLink SATCOM security enhancement. ARINC CDN (Content Delivery Network) protection. Ground network quantum hardening for ARINC 429/629/717 gateways.

Inmarsat and VSAT Providers: SwiftBroadband quantum-safe session protection. Classic Aero and Swift64 backward-compatible security. GX Aviation (Ka-band) full PQC implementation.

DEPLOYMENT TIMELINE AND PERFORMANCE

36-Week Deployment | 99.999% Availability | <50ms Authentication Latency

Phased Deployment Timeline

Phase 0	Safety Assessment and Gap Analysis (Weeks 1-4)
DO-326A threat assessment initiation. Protocol inventory and security gap analysis...	
Phase 1	Ground Infrastructure Deployment (Weeks 5-12)
QBITEL Bridge ground node deployment at primary ATC facilities. ADS-B authentication...	
Phase 2	Authentication Network Enrollment (Weeks 13-20)
Aircraft operator enrollment in authentication PKI. Ground receiver software update...	
Phase 3	AI Spoofing Detection Activation (Weeks 21-28)
Multilateration receiver network integration and calibration. AI model training ...	
Phase 4	Full Operational Status (Weeks 29-36)
Authentication verification mandatory for enrolled aircraft. Spoofing detection ...	

Performance Specifications

Authentication Performance

Metric	Specification	Notes
ADS-B Authentication Latency	<50 ms	End-to-end, ground receiver to ATC display
Message Throughput	500,000+ msg/hour	Per QBITEL Bridge node
False Positive Rate	<0.001%	Fewer than 1 in 100,000 authenticated messages
Spoofing Detection Rate	>99.7%	Systematic spoofing attacks
Ghost Aircraft Detection	<10 seconds	Time from injection to ATC alert
Position Falsification Detection	<30 seconds	Gradual position shift attacks
System Availability	99.999%	Five nines — matching ATC reliability standards

Automatic Failover Time	<500 ms	Failover to secondary node
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Cryptographic Algorithm Performance

Algorithm	Sign Time	Verify Time	Raw Size	Compressed
ML-DSA-65 (Dilithium-3)	<10 ms	<5 ms	3,309 B	659-1,317 B (60-80% reduction)
Falcon-512	<20 ms	<2 ms	897 B	359-538 B (40-60% reduction)
ML-KEM-768	<1 ms	<1 ms	1,088 B	N/A (KEM)
SLH-DSA-128f	<50 ms	<5 ms	17,088 B	6,835 B (stateless hash-based)
AES-256-GCM	<0.1 ms/KB	<0.1 ms/KB	N/A	N/A (symmetric)

COMPETITIVE DIFFERENTIATION

vs. Radar-Only | vs. ICAO Working Groups | vs. Classical Vendors | vs. IT-Adapted Solutions

vs. Radar-Only Security (SSR Mode S)

Argument: "We have secondary surveillance radar. We do not need ADS-B authentication."

Reality: SSR Mode S provides valuable cross-checking but has no global oceanic coverage, is subject to electronic warfare and interference, provides lower resolution altitude encoding, and does not protect commercial flight tracking aggregators used by airlines and passengers. QBITEL Bridge adds AI-driven multilateration cross-validation that correlates ADS-B with SSR data in real time — enhancing rather than replacing radar.

vs. ICAO Working Groups (Wait for Standardization)

Argument: "We will wait for ICAO to standardize ADS-B authentication."

Reality: ICAO ADS-B authentication working groups have been active for over a decade with no standardized solution deployed. Standardization timelines typically extend 5-10 years from initial proposal to mandated implementation. During that period, the threat environment continues to evolve and quantum computers advance. QBITEL Bridge provides operational security now while positioning customers for seamless compliance with future ICAO mandates. Early adopters gain operational experience and certified infrastructure when standards mandate implementation.

vs. Classical Security Vendors (RSA/ECDSA)

Argument: "Our current cryptographic systems are adequate."

Reality: RSA/ECDSA-based systems will be broken by quantum computers estimated 2030-2040. Aircraft deployed today will still be flying then. "Crypto agility" promises from legacy vendors require software updates — and in airborne systems, software updates require DO-178C recertification. QBITEL Bridge implements NIST-standardized PQC algorithms (FIPS 203, 204, 205) from day one, eliminating the need for an emergency cryptographic migration that would require recertification of deployed airborne systems.

vs. IT-Adapted Enterprise Security Solutions

Argument: "We can adapt our existing enterprise security tools for aviation."

Reality: General IT security vendors adapting enterprise products to aviation have no design experience for 600 bps bandwidth constraints, no understanding of DO-178C/DO-326A certification requirements, cannot generate aviation regulatory compliance evidence, have no experience integrating with ASTERIX, SDPS, and SWIM protocols, and have no operational precedent in safety-critical environments. QBITEL Bridge was architected for aviation from day one — not retrofitted from enterprise IT.

CUSTOMER SCENARIOS

Air Navigation Service Provider | Aircraft OEM | Airline CISO

Scenario 1 Air Navigation Service Provider (ANSP)

Profile: Major ANSP — 25+ ATC facilities, 3,000+ daily flights, NextGen/SESAR modernization

Challenge: 180K+ daily unauthenticated ADS-B messages, SWIM connections over internet infrastructure

QBITEL: ADS-B auth at 47 receiver sites, quantum-safe VPN mesh, AI spoofing detection, SOC

Outcome: National airspace ADS-B authentication within 6 months, <0.001% false positive rate

Timeline: 28 weeks to Full Operational Status | Contact: enterprise@qbitel.com

Scenario 2 Aircraft OEM — New Aircraft Program

Profile: Commercial OEM, new program, EIS 2027, service life to 2060-2065

Challenge: 40-year security horizon, DO-178C scope minimization, multi-authority evidence

QBITEL: ARINC 653 partition design from program inception, DAL-D DO-178C evidence package

Outcome: Security at certification baseline (no STC), PQC comms from EIS, 40-year roadmap

Timeline: Aligned with aircraft program schedule | Contact: certification@qbitel.com

Scenario 3 Airline CISO — Operations Security Program

Profile: 400+ aircraft, 120+ destinations, 6 continents, diverse fleet B737/777/787, A320/330/350

Challenge: FAA/EASA/CAA compliance, ACARS/SATCOM unsecured, harvest-now-decrypt-later threat

QBITEL: AOC security infrastructure, ACARS gateway auth, SATCOM session PQC, ADS-B monitoring

Outcome: All ACARS authenticated within 20 weeks, oceanic SATCOM quantum-safe, regulatory compliance

Timeline: 20 weeks | Contact: enterprise@qbitel.com

NEXT STEPS AND CONTACT

[Schedule a Briefing](#) | [Request Assessment](#) | [Initiate DO-326A Gap Analysis](#)

Why Act Now

The Quantum Clock Is Running: NIST finalized PQC standards August 2024. The U.S. government mandates PQC migration by 2035. Aircraft certified today will operate until 2060. The time to implement quantum-safe aviation security is at the beginning of new programs and network modernization initiatives — not when a regulatory mandate forces emergency implementation.

The Threat Is Present, Not Future: ADS-B spoofing with \$20 hardware is not theoretical. It is a documented, demonstrated, currently-achievable attack. Every day without authentication is a day the aviation system operates on trust rather than cryptographic verification.

Early Movers Gain Regulatory Advantage: Aviation regulatory processes reward early engagement. Organizations beginning DO-326A compliance documentation today will be positioned ahead of competitors when authorities mandate aviation cybersecurity certification. QBITEL Bridge provides the compliance evidence infrastructure to support that advantage.

Immediate Actions Available

For Air Navigation Service Providers: Schedule technical briefing | Request ADS-B vulnerability assessment | Initiate DO-326A gap analysis | Request non-operational demonstration

For Aircraft OEMs and Avionics Integrators: ARINC 653 partition architecture review | DO-178C evidence scope discussion | FAA/EASA Issue Paper strategy | Bandwidth optimization review

For Airlines: Airline CISO quantum threat briefing | ACARS/SATCOM security assessment | Regulatory compliance alignment review | Fleet security roadmap

For Military and Defense Aviation: Contact QBITEL defense aviation team | DoD PQC mandate alignment | ADS-B military airspace coexistence planning

Contact QBITEL Bridge Aviation Practice

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https://bridge.qbitel.com	DO-326A and DO-178C evidence support	24/7 aviation security monitoring

QBITEL Bridge — Securing Aviation's Future, Today

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